ChatGPT

2027 Raw Minerals Derivative Issuance: Executive Summary

In 2026–27 the global market for raw minerals (e.g. steel, copper, lithium) is robust, driven by energy transition and infrastructure demand. For example, global steel demand is forecast at ≈1.77 billion tonnes in 2026【58†L199-L203】, and World Bank analysts project a **17% jump** in a metals-and-minerals price index in 2026 owing to "steadily growing demand" and tight supplies【46†L970-L978】【46†L979-L987】. Base metals usage remains "structurally solid" with strong tailwinds from renewable energy, electrification and data centers【46†L979-L987】. Critical minerals for batteries also see surging forecasts – BloombergNEF expects copper demand to rise ~61% by 2050 (to ~46 Mt) and lithium demand to nearly triple (to 4.8 Mt LCE)【52†L298-L303】【52†L319-L324】. These trends indicate high aggregate demand (and price pressure) for raw minerals entering 2027.

Against this backdrop, a UK bank issuing a commodity-linked derivative in 2027 must navigate evolving UK/EU regulation and design structures that hedge producer costs while capturing commodity price upside. Key regulatory points include FCA and EMIR rules on commodity derivatives (position limits, clearing/margin, ancillary exemptions)【15†L310-L319】【26†L139-L144】. We propose and compare several structures (futures, options, swaps, structured notes, tolling/revenue-sharing) with detailed payoff and pricing models. Quantitative examples (cashflow projections, sensitivities) illustrate how each structure can reduce producers' marginal cost exposure and boost resale value. An exemplar term sheet and draft clauses outline legal documentation. Finally, we present an implementation roadmap, risk checklist, and KPIs to monitor performance and compliance.

1. Market Context (2026 Demand)

- **Base metals and steel**: World Bank (Apr 2026) forecasts a **17% rise** in its metals-and-minerals price index in 2026, reflecting strong industrial demand and supply tightness【46†L970-L978】. It notes that **base metals consumption remains "structurally solid"**, driven by traditional uses plus booming sectors (renewables, EVs, data centers)【46†L979-L987】. Indeed, long-run drivers are in place: copper, aluminium and tin prices are expected to hit record highs in 2026【46†L979-L987】, even as iron ore may dip (China steel demand weakness). For steel specifically, the World Steel Association projects *≈1.75–1.78 billion tonnes* global demand in 2025–26 (with ~1.773 bn t in 2026)【58†L199-L203】, driven by infrastructure and stabilized Chinese construction.

- **Critical/energy-transition minerals**: Forecasts for battery and strategic minerals are even steeper. BloombergNEF (Oct 2026) highlights that **copper demand** is set to climb 61% by 2050 (from ~28 Mt in 2026 to ~46 Mt)【52†L298-L303】, reflecting EV and grid growth. Lithium demand (battery-grade) is also surging: BNEF expects **~1.6 Mt LCE in 2026** climbing to 4.8 Mt by 2050【52†L319-L324】. (Cobalt/graphite similarly rise, though mitigated by battery tech shifts.) Precious metals (gold, silver, platinum) are at multi-year highs on investment demand and supply constraints【46†L998-L1007】.

- **Aggregate raw material outlook**: These indicators signal robust aggregate demand for raw minerals in 2026, supporting high price baselines as of early 2027. Supply-side bottlenecks (geopolitics, infrastructure limits) mean price volatility remains a concern. A UK bank issuing a

commodity-derivative linked to such minerals would thus operate in a **strong-demand, high-price environment**, which shapes the design goals: locking in favourable terms to lower producers' marginal costs today, while preserving upside if markets rally further.

2. Regulatory & Legal Constraints (UK/EU)

(a) UK Regulatory Framework: UK firms must comply with Financial Conduct Authority (FCA) and Prudential Regulation Authority (PRA) rules. Since Brexit, UK transposed MiFID II, EMIR etc. but has reformed some commodity-derivatives rules. Under UK law (FSMA 2000 and the Regulated Activities Order), dealing or arranging commodity derivatives is a regulated activity, unless an **Ancillary Activities Exemption (AAE)** applies【26†L139-L144】. In practice, only commodity firms with trading "ancillary" to a non-financial business can use the AAE test. Notably, HM Treasury has **simplified the AAE**: from Jan 2025 UK firms no longer must notify the FCA annually of AAE status【26†L139-L144】. Until a new regime is implemented (target Jan 2027), firms can rely on the existing quantitative test【26†L139-L144】. This means a bank (as a financial firm) writing a commodity derivative would itself be authorised (the AAE would not exempt it), whereas a non-financial miner might still use the AAE.

- **Position limits and reporting:** The FCA has adopted MiFID II-style limits on commodity derivative positions. Final rules (FCA PS25/1, Feb 2025) apply from 6 July 2026【15†L338-L347】. New UK rules focus position limits only on "critical contracts" (energy and base metals) and their economically related contracts, whereas EU MiFID II limits apply to agricultural and "significant" commodities. The UK also shifted setting of limits to trading venues (exchanges) rather than FCA alone【19†L195-L203】. Exemptions are expanded in the UK: e.g. (i) financial firms hedging the commercial risks of non-financial clients, and (ii) liquidity providers meeting venue rules【19†L206-L209】. Trading venues must also set accountability thresholds beyond limits【19†L210-L213】. Banking groups issuing derivatives will need to monitor aggregate positions (own and clients') to ensure compliance.

- **Trading and clearing:** Many commodity derivatives (especially liquids like oil, gas, metals futures) trade on exchanges and use clearinghouses. By 2027, UK EMIR-like rules will mandate central clearing for certain OTC commodity swaps and futures with "clearing obligation". Variation and initial margin requirements apply to non-cleared OTC trades under UK EMIR (mirroring EU). In addition, trades must comply with Market Abuse Regulation (MAR) – e.g. spot or derivative markets cannot be manipulated (especially important in physical-settlement terms). A UK bank must use authorised venues or MTFs unless a MiFID exemption applies (e.g. professional clients only). If targeting EU investors, EU MiFID II/MiFIR and EMIR rules (e.g. EU equivalence or substitute compliance) may also apply.

- **Capital and business standards:** As a bank, issuing derivatives falls under CRR/CRD IV capital and liquidity regimes (or IFR for investment-firm subsidiaries). Counterparty credit risk exposures (CVAs) and market risk under FRTB must be capitalized. Internal risk models (IRB/AMA) may cover commodity stresses. UK rules also impose "prudential trading book" standards for commodity inventories or market-making exposures. The PRA would require robust risk systems for market, credit, liquidity and operational risk on commodity desks.

- **Legal documentation & tax:** Transactions will use standard contract forms (e.g. ISDA 2022 Commodity Definitions for OTC derivatives). Sales of structured notes fall under FCA Prospectus or MiFID product rules if offered to UK/EU retail; for professional issuance, documentation must meet MiFID MiFID II disclosures. Taxes: commodity derivatives are generally exempt from Stamp Duty (SDRT) and VAT in the UK, as they are financial instruments. However, careful tax treatment

is needed: e.g. structured-note coupons count as interest/dividends (subject to withholding tax if foreign payor), and any physical settlement triggers standard sales tax/VAT on the commodity. Issuers must also consider withholding on any embedded debt. Transfer pricing and CFC rules rarely apply unless group non-UK entities are involved.

- **Industry guidance:** The FCA and ESMA have issued warnings on commodity trading conduct (e.g. insider dealing, attempted market cornering in energy). By 2027, firms will be expected to follow enhanced surveillance, and possibly new UK/EU guidance (e.g. on position reporting quality). The bank must also follow general conduct rules (fair dealing, best execution if client trades, anti-money laundering checks on mining counterparties, etc.).

3. Proposed Derivative Structures

We consider several derivative structures to achieve the goals (lower marginal cost, capture upside):

- **Futures/Forwards (Linear Hedging):** A simple forward/futures contract locks in a sale price. *Payoff:* linear (P_fixed–spot or spot–P_fixed). *Model:* Forward price $F = S_0 \cdot \exp((r+\text{carry}-\text{storage})\cdot T)$. Pricing via cost-of-carry. *Hedging:* Delta=–1 (producer would sell forward; bank takes opposite side). *Pros/Cons:* Eliminates price risk (guarantees sale price) but forfeits any upside beyond that price. Margin and collateral per exchange/EMIR. Counterparty risk is minimal if exchange-cleared; OTC forwards would need CSA/IM per regulation. *Tax:* no SDRT/VAT; gains taxed as income/profit.

- **Options (Asymmetric Payoff):** We can structure **puts** or **calls** (or combinations, e.g. collars). *Example:* Producer buys a put (strike K) to lock in a floor price, or bank sells a covered call on behalf of producer to collect premium. *Payoff:* e.g. long put gives max(K–S_T,0) – premium. *Pricing:* Black (1976) or Black's (1976) models for commodity options, calibrated to implied vol. *Hedge:* Delta-hedge the option with underlying commodity or futures. *Advantages:* Puts/collars ensure a minimum sale price (protecting marginal cost) while still allowing participation in high prices (above strike). *Drawback:* premium cost; complex hedge. Exchange-traded commodity options exist (e.g. on futures), or OTC options (clearing/margin or bilateral). *Counterparty:* If OTC, margin under EMIR/UK EMIR; exchange options have clearing. *Tax:* Option premiums paid are tax-deductible; gains on exercise follow same tax treatment as underlying sale.

- **Commodity Swap:** A fixed-for-floating swap (e.g. bank pays fixed price and receives floating commodity price each period). *Payoff:* Like a series of forward contracts: net cashflow = (Spot_t – P_fixed). *Pricing:* Set P_fixed = forward curve (mid-market swap rate). *Use:* Can smooth margin by receiving variable revenue if spot is high (as swap receiver), or paying fixed price to hedge (if swap buyer). *Hedge:* Replicate via forwards/futures stacks. *Collateral:* Usually OTC swap -> initial + variation margin. *Pros:* Custom tenor and notional; *Cons:* linear like forward (no upside beyond fixed price).

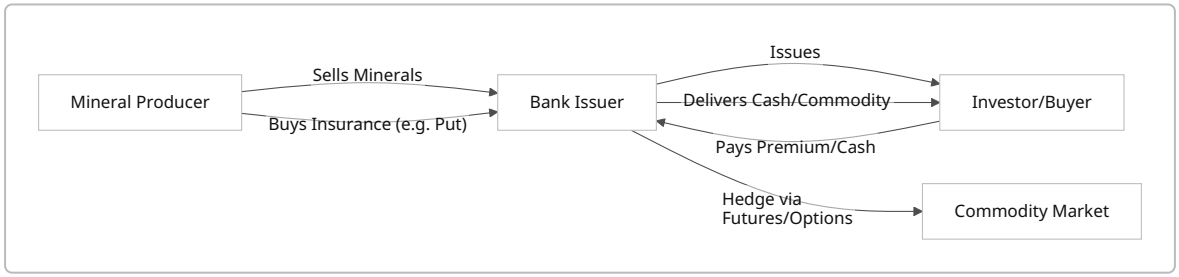
- **Structured Note (Commodity-Linked Bond):** A bond or note whose payoff includes commodity-indexed components. For example, a **note with a call spread**: investors receive fixed interest plus a cash payoff = Notional·max(min(S_T,K_hi)–K_lo,0). *Payoff diagram:* Piecewise linear (zero below K_lo, capped above K_hi). *Pricing:* Discounted cashflow + Black/Bachelier pricing for embedded options. *Hedge:* Bank hedges option component via dynamic delta-hedging. *Counterparty:* The bank (issuer) bears risk until hedged; investor counterparty is typically a financial or corporate buyer. *Collateral:* If OTC, similar to swaps; if note is publicly issued,

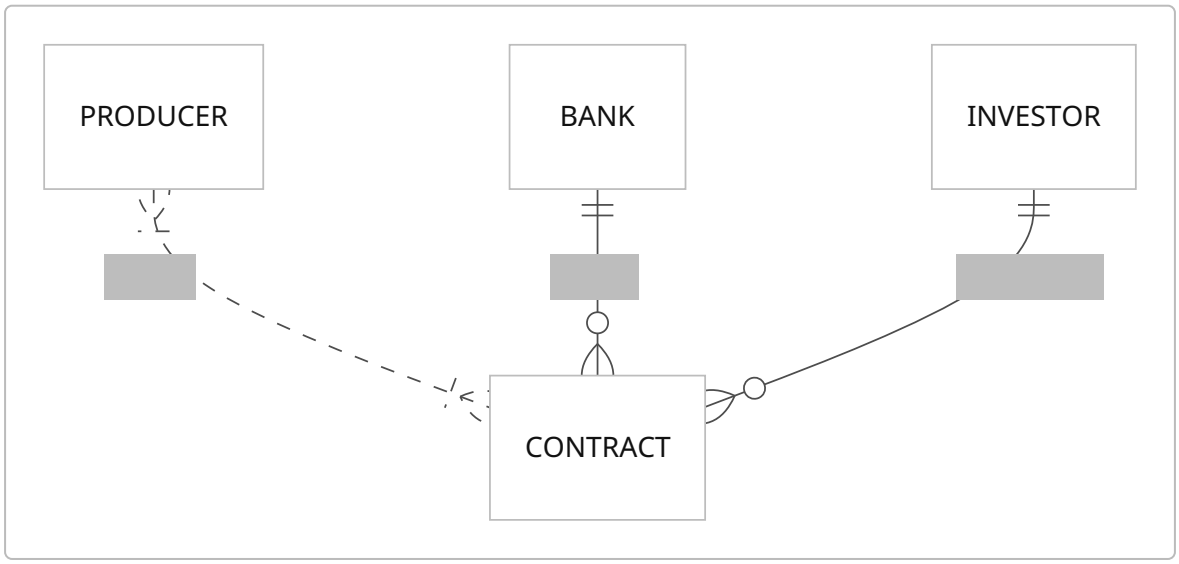
collateral less relevant but bank must manage embedded risk. *Tax:* Coupons taxed as interest; any commodity payoff taxed as capital gains or trading profit.

- **Revenue-Sharing/Offtake Agreement (Stream):** The bank provides financing or guarantees and receives a share of mineral revenues. *Structure:* Could be a **streaming** contract where the bank pays an upfront or periodic fee, then receives a fixed fraction (e.g. 50%) of the producer's mineral output at market price (or at a fixed discount). *Pricing:* Present value of expected cashflows (depends on volume forecasts and price distribution) or from simulation. *Hedge:* The bank's return is directly linked to mineral prices; to lower producers' cost, the bank may pay a fixed cost or provide services (like tolling below). *Counterparty Risk:* Structured like a joint venture; subject to mining project risk (political, operational). *Tax:* Often treated as service or royalty income – may attract VAT or royalty-like taxes.

- **Tolling Agreement:** Common in energy/mining: the bank finances processing facilities (or pays fixed feedstock cost) in exchange for refined output. *Mechanics:* Producer delivers raw ore to toller (bank's facility) and pays a tolling fee (often fixed). The bank sells the refined product and splits revenue with producer (or keeps excess). *Effect:* Lowers producer's effective cost of processing (the fee instead of variable cost) while giving bank upside if product prices rise. *Legal Form:* Could be structured as an asset-backed financing with embedded commodity swap (synthetic toll). *Tax:* The toll fee might be expense-deductible for producer; bank income subject to corporate tax.

Payoff Diagrams: Below are schematic payoffs for selected structures (illustrative only):





Counterparty & Collateral: All OTC derivatives (swaps, options, structured notes) will typically be documented under an ISDA Master Agreement (with an appropriate Commodity Annex). Bank's

derivative trading desk must manage counterparty credit via CSAs: initial margin for non-cleared trades, variation margin daily. For listed futures/options, the clearinghouse mitigates risk (collecting margin, default funds). Counterparty risk to the bank arises if clients (or a trading counterparty) default; mitigations include netting agreements and requiring cash or liquid collateral. Liquidity risk is also present: in stress, markets can gap, so clients must provide sufficient margin or the bank must repo risk.

Tax Implications: Derivative gains are generally taxed as trading income to the producer (deductible as production cost hedges). If derivatives are part of a financing (e.g. structured note), any interest coupon is subject to income tax rules. VAT: purely financial derivatives (swaps, forwards, options) are VAT-exempt. For tolling/streaming, the "service fee" might attract VAT. UK "Furnished Holiday Let" rules or transfer pricing can apply if the bank's structure is interpreted as part of a broader financing arrangement; careful documentation is needed to avoid unexpected tax on commodity flows.

4. Quantitative Modelling & Cashflows

To illustrate, assume a mineral producer extracting **50,000 tons** of "Metal X" annually. Current production cost is £40/ton (variable), and the spot price is £60/ton. We compare three hedging strategies and an unhedged baseline:

Strategy	Hedge Instrument	Effective Sale Price	Profit Margin
No hedge	–	Future spot	(P–40)
Forward Sell	Sell forward at £55	55 (locked)	15 per ton
Buy Put	Long Put (K=50, premium £2)	≥50 (floor at 50–2)	10 (floor) to upside
Collar	Long Put K=50, Short Call K=70	50–70 range	10–28 per ton (depends on spot)

Assumptions: forward curve implied ~55, put option 1-year strike 50 costs £2/ton premium, call strike 70 receives £2.

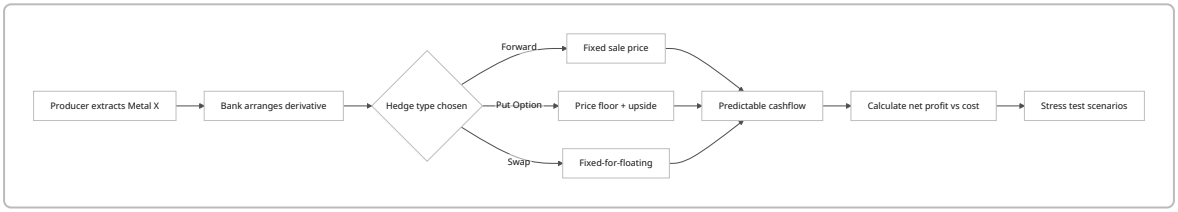
- **Cashflows:** For each strategy we compute yearly cashflows across price scenarios (£/ton): 30, 50, 60, 80. (See Table below.)

Price P	No Hedge Margin	Forward at 55	Put K=50	Collar (Put50+Call70)
30	–10 (loss)	15	8	8 (–10+2)
50	10	15	8 (50–40–2)	8 (–)
60	20	15	18 (60–40–2)	18 (–)
80	40	15	38	18 (capped at 70–40+2)

- **Interpretation:** The *forward* (locked at 55) guarantees a margin of +15/ton regardless of spot, eliminating downside but forgoing upside. The *put* hedge ensures a **floor** (~£8 margin) when prices fall, while letting margins rise with spot (minus £2 premium). The *collar* (short call) yields

similar floor but caps maximum margin at £18. The unhedged producer bears full price swings (from –£10 to +£40 per ton in our table).

- **Optimisation:** The bank can use these projections to "optimise" parameters (strike, hedge ratio) to minimize *marginal cost of production*. For example, choosing a put strike ≈ expected cost level locks minimum profitability. Sensitivity analysis (below) quantifies how net profit varies with price or volume assumptions. Stress tests should include extreme price drops (e.g. P<£30) or production shortfalls (e.g. 20% lower output).



Cashflow Chart (Example): The chart below shows net profit (margin) versus spot price for the no-hedge vs put-hedge scenarios (profit per ton):

```
```{="chart"}

{
  "type": "line",
  "data": {
    "labels": [30,50,60,80,100],
    "datasets": [
      {"label": "No Hedge", "data": [-10,10,20,40,60], "borderColor":
"blue"},
      {"label": "Put (K=50)", "data": [8,8,18,38,58], "borderColor": "green"}
    ]
  },
  "options": {"title": {"text": "Producer Profit vs Spot Price", "display":
true}}
}
```

(In the chart, "No Hedge" is in blue; "Put (K=50)" in green. The hedge provides a flat floor at ~£8, whereas unhedged profit rises linearly.)

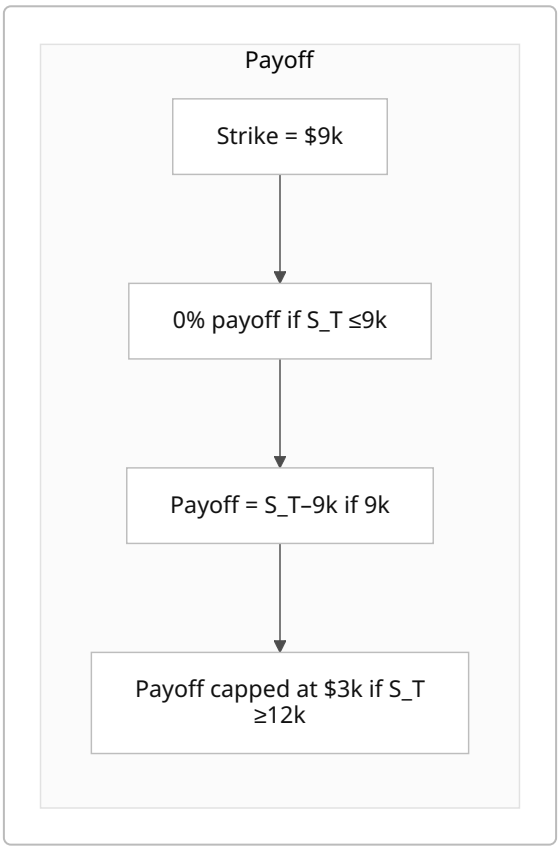
Sensitivity & Optimisation: The bank can build a simple model (e.g. in Excel) where output volume and commodity price are stochastic (e.g. normal or scenario-based). By adjusting hedge ratios or strike levels, it can minimize profit variance subject to a target expected margin. For example, solving Minimize Var(profit) subject to E(profit)≥10 yields an optimal mix of forwards and options. These optimisations would use historical volatilities (for BS model) and correlations between costs and prices.

5. Exemplar Term Sheet & Legal Clauses

Term Sheet (illustrative):

- **Issuer:** ABC Bank (UK) – credit-rated, regulated by FCA/PRA.
```

- **Buyers:** Qualified institutional investors (e.g. mining companies, traders).
- **Underlying:** 100,000 tonnes of Copper (LME Grade A, spot contract).
- **Instrument:** 5-year commodity-linked note (2027–2032), cash-settled.
- **Notional:** 100,000 t × $10,000/t (notional value $1.0 bn).
- **Payoff:** At maturity, the note pays a coupon = Notional × max(0, S_T – K), where S_T is the LME copper price on 1 Jan 2032, and strike K=$9,000/t. (This replicates a digital call above 0.) In addition, a fixed coupon of 3% p.a. on notional is paid semi-annually.
- **Strike/Trigger:** $9,000/t, with cap at $12,000/t (see payoff diagram below).
- **Settlement:** Cash in USD on 15 Jan 2032 (3 business days after valuation). No physical delivery.
- **Margin:** Issuer to post collateral to cover mark-to-market; Investor may post upfront deposit of 5%. Daily MTM netting.
- **Governing Law:** English law; disputes via London arbitration.



Key Draft Clauses (samples):

- **Payment (Settlement):** “On the Maturity Date, the Issuer shall pay the Investor (on a cash settlement basis) an amount equal to `Notional × max(0, S_T – K)`, subject to any cap. All calculations use the official LME spot price at the Valuation Time. Payment shall be made in USD on the Payment Date.”

- **Events of Default:** “Standard ISDA default events: failure to pay or deliver, insolvency, breach of a material term (e.g. unauthorized netting). Upon default, the non-defaulting party may close out positions and net obligations. For example, if ABC Bank becomes insolvent, the Investor may terminate the note and claim any positive market value (to the extent recoverable from administration).”

- **Force Majeure:** “If a Force Majeure Event (e.g. war, embargo on metals, natural disaster) makes settlement impossible, the calculation agent may postpone the Valuation Date or replace S_T with an industry-consensus proxy price. Neither party is liable for non-performance if excused by force majeure. Following such an event lasting >30 days, either party may elect early termination at a reasonable market value.”

- **Settlement Mechanism:** “If cash settlement fails (e.g. missing price), fallback to the nearest liquid contract (e.g. 3-month swap) or governing benchmark. Delivery on London business days. Business Day: London business day for LME. Calculation agent: XYZ Bank (an affiliate of ABC) will compute the payoffs; in case of dispute, parties use ISDA methods (good faith, no arbitrage).”

- **Governing Law and Jurisdiction:** “This instrument is governed by English law. Any disputes shall be resolved by arbitration in London under LCIA rules. Consent to jurisdiction (if needed).”

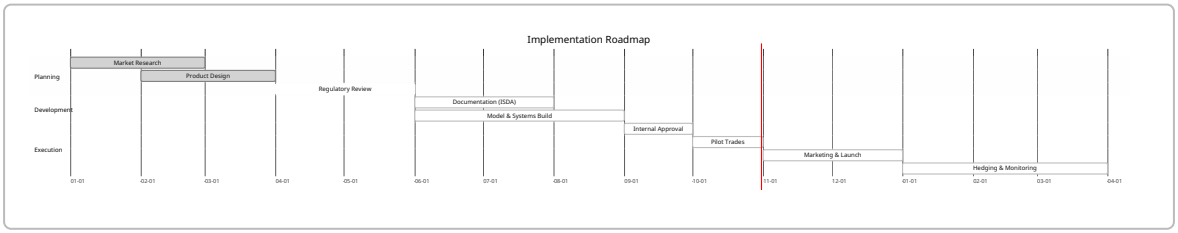
- **Tax and Gross-Up:** “Payments are made net of withholding taxes. If the Issuer is required by law to deduct tax, the Investor will receive a grossed-up amount to achieve the same net payment. Both parties represent tax residency and compliance with FATCA/CRS.”

(These terms follow typical ISDA-style provisions for commodity swaps, adapted to a public note.)

6. Implementation Roadmap & Risk Management

An efficient implementation requires coordination across front-office, risk, legal, IT and compliance. Key steps include:

- **Q1–Q2 2026: Feasibility & Design.** Conduct market research (demand/pricing). Define product features (type, tenor, notional). Assess regulatory requirements (e.g. FCA approvals, CRD allocation). Draft term sheet and obtain internal approvals.
- **Q2–Q3 2026: Documentation & Setup.** Prepare legal documents (ISDA, bank-bond docs, prospectus if public). Set up risk management systems: define hedge strategy, counterparty limits, margining framework. Develop and test pricing/valuation models (Black/BS vs real path simulation). Coordinate with compliance on marketing materials (financial promotion rules).
- **Q3–Q4 2026: Approvals & Pilot.** Engage with regulators (FCA notifications on new product lines). Perform senior management and credit committee approvals. Pilot trade(s) with affiliate or selected counterparty to debug processes. Train sales and risk staff on product details and systems (P&L booking, collateral mgmt).
- **Late 2026 – Early 2027: Launch & Execution.** Market the offering to targeted clients (miners, funds). Execute hedging transactions (e.g. execute futures/options to offset exposure). Manage collateral accounts and daily MTM. Perform real-time monitoring of compliance (e.g. limits, position reports). Close issuance and begin ongoing client reporting.



Risk Management Checklist:

- **Market Risk:** Set position and sensitivity limits (deltas, convexity). Use stress tests (extreme price drops/rallies). Hedge residual exposures (e.g. cross-commodity or FX).
- **Credit/Counterparty Risk:** Evaluate clients' creditworthiness. Require collateral via CSA. Use clearinghouses for standardized trades. Monitor issuer's own credit (ISDA cross-default).
- **Operational Risk:** Ensure correct trade booking (commodity IDs, margin calls). Perform middle-office reconciliations daily. Plan for business continuity (backup feed for price, alternate valuations).
- **Liquidity Risk:** Keep liquid instruments for hedging. Maintain committed lines or reserves in case of massive margin calls or market freeze.
- **Legal/Compliance:** Verify documentation (no veto mismatches). Check KYC on counterparties (AML for mining business). Ensure product governance (suitability rules, any disclosure to clients).
- **Reputational/Strategic:** Align issuance with bank's risk appetite. Avoid overly complex or opaque structures that could draw regulatory scrutiny.

Each risk area should have designated ownership (e.g. CRO oversight, trading limits in front office, compliance monitoring), with periodic review (monthly reporting to ALCO or risk committee).

7. KPIs and Monitoring Metrics

To track the success and safety of the issuance, we recommend the following KPIs:

- **Hedging Effectiveness:** Variance of realized margin vs unhedged case. (Target: >70% variance reduction.)
- **Coverage Ratio:** Volume hedged ÷ production volume. (Target: e.g. 80–100% cover.)
- **Mark-to-Market Exposure:** Daily MTM of derivative portfolio; should stay within limits (e.g. $\leq$X% of Tier 1 capital).
- **Counterparty Concentration:** Largest counterparty exposure as % of credit limit. (Maintain low concentration.)
- **Margin Utilization:** Percentage of margin calls met on time; cash/eligible collateral held vs required.
- **Regulatory Compliance:** Timeliness of position reports (per FCA/ESMA) and any limit breaches. (Zero breaches expected.)
- **Financial KPIs:** Net Profit from the transaction (vs budget); IRR on issued note; Carry cost vs realized price (margins).
- **Operational Metrics:** Trade confirmation errors (target 0), settlement failures, audit findings.

These metrics should be reported monthly to management. Deviations (e.g. price movements causing large MTM losses) should trigger an immediate review and possible strategy adjustment.

Sources: Market demand and price trends are drawn from industry reports and regulators【46†L970-L978】【58†L199-L203】【52†L298-L303】. Regulatory requirements are based on UK FCA Policy Statements and legislation【15†L310-L319】【26†L139-L144】. Structures and modelling are standard in fixed-income/commodities finance (see e.g. BloombergNEF or ISDA literature for pricing).
